

1. **Experiment:** AlphaFold3 Prediction

2. **Time:** 2025.11.23-2025.12.01

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4. **Method:**

(1) Data Retrieval

- ① Retrieve Template Structures: The model retrieves template structures from PDBx/mmCIF files. This involves parsing the files to extract the relevant structural information that can serve as a template for the prediction of the target protein structure.
- ② Database Sequence Retrieval: Retrieve protein sequences from five databases to be used for multiple sequence alignment (MSA). And retrieve RNA sequences from three databases, also for MSA purposes. These sequences help in understanding the evolutionary context and the potential interactions with the protein of interest.

(2) Sequence and Token Processing

- ① Tokenization: Convert the input sequences (protein and RNA), ligands, and covalent bonds into tokens. Tokenization is the process of breaking the sequences into smaller units that can be processed by the neural network.
- ② Embedding: Input the tokens into an embedder, which is a component of the neural network designed to convert tokens into high-dimensional vectors (embeddings) that capture their meaning in the context of the model. The embedder will output three types of embeddings:
Input Embeddings: Representations of the input sequences.
Single Embeddings: Representations of individual tokens.
Pair Embeddings: Representations of pairs of tokens, which capture the relationships between different parts of the sequences.
- ③ AF3 prediction: Input sequences are submitted to the AF3 web server. Altering the order of the three input sequences produces varied protein-structure outputs and different pTM and ipTM scores.

(3) Structure Generation and Refinement

- ① Integration of Information: Combine the template structure information with the pair embeddings obtained from the feature extraction module. Integrate MSA information to generate a pair representation that captures the evolutionary and structural relationships within the sequences.
- ② Representation Regulation: Use the Pairformer module to organize and update the single representation and pair representation information. This module ensures that the relationships between different parts of the protein are accurately reflected in the model's representations. And the single and pair representations are updated iteratively through multiple cycles to refine and improve their accuracy.
- ③ Structure Prediction: Import the resulting protein files into PyMOL, label distinct protein chains, and sequentially execute scripts for hydrogen-bond analysis, electrostatic-interaction analysis, interface flexibility analysis and BSA calculation to obtain the corresponding data.

5. **Result:**

To investigate the stability of the bound complexes, we performed 90 replicate AlphaFold3 predictions for three candidate combinations and their binding models with the IL-6/sIL-6R α complex (Extended Data Tables 1-2). The five predictions with the highest ipTM scores were selected as high-confidence models for in silico complex construction (Extended Data Table 3). A2B3C3 (candidate 1) exhibited a larger buried surface area than A3B2C2 (candidate 2) (1801.790 vs. 1759.434 Å²). Interface flexibility assessed by pLDDT indicated that the A2B3C3-sIL-6R interface was relatively flexible, whereas the A2B3C3-IL-6, A3B2C2-IL-6, and A3B2C2-sIL-6R interfaces were relatively rigid (Extended Data Tables 1-2). These results suggest that A2B3C3 may first engage IL-6 through a rigid interaction, followed by an allosterically induced fit with sIL-6R, thereby enhancing

overall complex stability. We further performed alanine scanning to identify hotspot residues at the binding interfaces of the modeled complexes (Figure 1e; Extended Data Table 4). IL-6 Tyr13 and sIL-6R Trp154, which are hotspot residues involved in gp130 binding by the IL-6 trans-signaling complex, were also occluded by A2B3C3. Thus, the A2B3C3-bound complex is unlikely to retain the capacity to bind gp130.

Extended Data Table 1. 90 repeated AlphaFold3 predictions of the A2B3C3 virtual complex.

	ipTM	pTM	BSA (Å ²)	Hydrogen bond analysis		Hydrophobic pairing		Interface pLDDT	
				effective bonds	average energy	mimic with IL-6	effective bonds	average energy	mimic with sIL-6R
A2B3C3_1	0.62	0.66	1829.733	14	-19.71	17	14	75.7	49.4
A2B3C3_2	0.62	0.65	1877.327	11	-14.33	18	14	75.6	41.3
A2B3C3_3	0.63	0.67	1834.68	16	-19.48	17	21	75	45.4
A2B3C3_4	0.62	0.66	1573.99	14	-21.46	18	19	74.1	48.3
A2B3C3_5	0.63	0.68	1834.834	14	-17.01	18	23	75.3	45.7
A2B3C3_6	0.66	0.67	1999.601	14	-15.65	17	14	75.3	46.1
A2B3C3_7	0.59	0.65	1831.701	14	-19.08	18	8	74.4	39.6
A2B3C3_8	0.63	0.66	1684.835	14	-11.73	17	14	75.2	46.2
A2B3C3_9	0.65	0.66	1938.262	9	-8.65	18	16	74.7	43.4
A2B3C3_10	0.62	0.66	2433.042	19	-21.07	20	19	72.4	40
A2B3C3_11	0.63	0.67	1788.008	13	-15.53	18	21	75.6	45.8
A2B3C3_12	0.65	0.66	1628.776	11	-14.18	18	10	76.1	42
A2B3C3_13	0.64	0.66	2002.86	10	-11.51	19	6	75.3	39.5
A2B3C3_14	0.65	0.67	1871.37	11	-14.51	18	16	74.9	61.8
A2B3C3_15	0.64	0.68	1522.365	12	-14.31	18	5	76.1	56.7
A2B3C3_16	0.66	0.67	2190.491	10	-9.01	18	12	75.5	43.6
A2B3C3_17	0.63	0.65	1653.021	11	-17.26	18	18	74.5	34.1
A2B3C3_18	0.63	0.66	2222.7	10	-10.95	18	16	70.3	41
A2B3C3_19	0.62	0.66	1408.613	13	-13.29	18	15	75.1	47.5
A2B3C3_20	0.64	0.66	2224.081	17	-18.53	17	19	73.2	45.5
A2B3C3_21	0.63	0.66	1772.101	14	-19.6	18	10	76.3	41.3
A2B3C3_22	0.66	0.66	2309.83	13	-12.25	18	10	75.4	44.7
A2B3C3_23	0.62	0.67	1758.063	14	-13.73	18	26	75.4	42.8
A2B3C3_24	0.64	0.67	1989.224	9	-11.73	17	17	75	45.3
A2B3C3_25	0.64	0.67	1993.652	9	-10.44	18	18	75.2	52.6
A2B3C3_26	0.62	0.67	2037.689	19	-17.3	19	24	75	42.6
A2B3C3_27	0.63	0.65	1761.003	12	-8.27	17	21	75.1	46.1
A2B3C3_28	0.58	0.65	1808.27	15	-14.35	19	3	73.7	49.5
A2B3C3_29	0.65	0.66	2543.582	15	-14.26	19	26	70.2	43.2
A2B3C3_30	0.62	0.66	1588.402	16	-17.1	17	15	74.3	46.2
A2B3C3_31	0.62	0.66	1776.255	14	-16.2	19	18	76	57.6
A2B3C3_32	0.65	0.67	2008.63	15	-19.05	18	31	77.7	40.1
A2B3C3_33	0.64	0.67	1867.761	12	-16.3	18	9	74.5	56.4
A2B3C3_34	0.65	0.67	2151.016	10	-15.38	19	8	76.4	41.5
A2B3C3_35	0.64	0.66	2428.659	14	-18.06	19	16	66	43.2
A2B3C3_36	0.64	0.65	1650.311	16	-17.95	19	16	75.5	41.4

A2B3C3_37	0.63	0.66	2281.445	13	-15.02	18	11	74.8	42.3
A2B3C3_38	0.65	0.66	1658.327	13	-15.29	18	11	74.2	51.4
A2B3C3_39	0.59	0.67	2358.384	20	-26.31	18	31	65.3	43.2
A2B3C3_40	0.63	0.67	1769.669	15	-18.11	17	18	75.2	44.1
A2B3C3_41	0.62	0.71	1828.411	17	-21.04	18	26	75.5	70.2
A2B3C3_42	0.65	0.67	2064.925	14	-17.97	18	15	75.4	41.6
A2B3C3_43	0.63	0.67	1555.423	18	-24.03	18	23	74.6	44.1
A2B3C3_44	0.63	0.67	2247.854	15	-17.01	18	12	74.6	42.6
A2B3C3_45	0.67	0.71	1771.375	16	-24.18	19	15	76.5	67.6
A2B3C3_46	0.64	0.66	2356.609	10	-15.89	16	13	74	42.7
A2B3C3_47	0.64	0.66	1506.355	18	-25.43	18	18	75.6	54.4
A2B3C3_48	0.68	0.68	1626.115	19	-16.76	18	27	77	51.5
A2B3C3_49	0.64	0.67	1938.033	13	-19.6	16	22	76	43.2
A2B3C3_50	0.65	0.67	1373.333	9	-11.08	18	13	74.4	45.3
A2B3C3_51	0.64	0.68	9314.033	13	-14.3	18	21	75.1	46.8
A2B3C3_52	0.63	0.66	1601.409	10	-14.17	18	7	75.7	51.2
A2B3C3_53	0.63	0.67	2201.611	12	-18.29	17	22	75.2	41.6
A2B3C3_54	0.62	0.66	1827.47	18	-20.76	18	34	76	46.5
A2B3C3_55	0.62	0.66	1768.302	18	-21.54	19	18	75.5	39.7
A2B3C3_56	0.6	0.66	1797.276	14	-18.31	18	13	74.8	55.1
A2B3C3_57	0.62	0.66	1964.762	11	-16.37	18	10	74.6	38.6
A2B3C3_58	0.63	0.67	1642.997	13	-14.36	19	25	75.5	39.5
A2B3C3_59	0.64	0.67	1722.606	8	-8.09	19	13	76.1	68.8
A2B3C3_60	0.61	0.67	1606.459	10	-10.4	18	20	74.8	42
A2B3C3_61	0.63	0.67	1791.736	12	-9.37	17	32	66.66	40.81
A2B3C3_62	0.64	0.66	1884.295	17	-21.53	17	15	75.21	63.42
A2B3C3_63	0.63	0.67	1809.749	14	-12.63	18	15	71.47	55.07
A2B3C3_64	0.65	0.67	1902.02	11	-21.67	19	18	67.74	50.23
A2B3C3_65	0.61	0.66	1999.601	11	-10.8	18	32	76.16	62.91
A2B3C3_66	0.62	0.65	1798.491	9	-23.14	18	11	68.63	53.75
A2B3C3_67	0.62	0.66	1798.493	15	-11.9	19	25	68.24	49.2
A2B3C3_68	0.65	0.67	1999.601	11	-9.23	19	9	69.92	34.79
A2B3C3_69	0.65	0.66	1916.078	17	-18.13	18	24	70.09	58.78
A2B3C3_70	0.64	0.66	1770.864	15	-19.3	16	9	68.19	51.72
A2B3C3_71	0.62	0.67	1889.678	16	-19.8	20	24	67.48	42.06
A2B3C3_72	0.62	0.66	1771.575	17	-22.79	19	15	70.24	53.97
A2B3C3_73	0.62	0.67	1771.304	18	-24.48	17	30	75.74	52.97
A2B3C3_74	0.65	0.66	1854.387	18	-17.52	17	10	75.06	51.7
A2B3C3_75	0.63	0.66	1601.36	16	-15.13	20	20	73.25	43.6
A2B3C3_76	0.63	0.67	1623.474	15	-8.98	19	13	71.45	53.14
A2B3C3_77	0.64	0.66	1759.968	14	-19.96	20	18	71.17	69.64
A2B3C3_78	0.64	0.68	1707.074	12	-21.27	20	8	71.18	56.61
A2B3C3_79	0.65	0.66	1862.874	8	-18.17	20	7	77	56.83
A2B3C3_80	0.64	0.66	1719.378	15	-8.86	19	23	74.45	66.95
A2B3C3_81	0.62	0.66	1660.175	16	-23.04	19	12	65.82	68.85
A2B3C3_82	0.64	0.66	1998.049	8	-23.71	16	12	74.7	55.22

A2B3C3_83	0.64	0.66	1799.475	19	-24.79	20	32	71.67	40.05
A2B3C3_84	0.65	0.68	1833.909	15	-16.34	19	29	67.51	48.24
A2B3C3_85	0.64	0.67	1658.714	18	-20.66	16	3	70.16	68.2
A2B3C3_86	0.63	0.66	1762.07	14	-21.16	18	31	76.95	46.5
A2B3C3_87	0.65	0.67	1839.003	11	-9.53	18	33	67.94	59.55
A2B3C3_88	0.61	0.67	1690.853	16	-18.41	19	25	75.38	50.81
A2B3C3_89	0.64	0.66	1870.088	13	-18.68	18	5	70.31	48.06
A2B3C3_90	0.65	0.68	1755.465	13	-8.34	19	15	72.58	53.18

Extended Data Table 2. 90 repeated AlphaFold3 predictions of the A2B3C2 virtual complex.									
	ipTM	pTM	BSA (Å²)	Hydrogen bond analysis		Hydrophobic pairing		Interface pLDDT	
				effective bonds	average energy	mimic with IL-6	effective bonds	average energy	mimic with IL-6
A2B3C2_1	0.62	0.71	1473.948	19	-25.63	16	10	72.4	70.4
A2B3C2_2	0.6	0.7	1402.993	18	-26.41	18	10	72.3	68.6
A2B3C2_3	0.62	0.71	1367.934	17	-27.36	20	9	71.4	71.1
A2B3C2_4	0.6	0.69	1720.724	15	-18.22	18	11	71.6	68
A2B3C2_5	0.62	0.71	1771.421	16	-19.57	19	10	71.2	71.9
A2B3C2_6	0.6	0.69	1354.058	18	-25.7	19	9	72.9	69.4
A2B3C2_7	0.62	0.71	1431.637	20	-29.53	18	11	71.4	71.6
A2B3C2_8	0.62	0.71	1581.135	17	-25.74	18	11	71.4	70.2
A2B3C2_9	0.6	0.69	1623.096	16	-19.82	20	11	73.9	68
A2B3C2_10	0.6	0.69	1442.243	18	-23.09	18	11	73.4	68.9
A2B3C2_11	0.62	0.71	1295.91	17	-24.12	17	10	71.3	70.4
A2B3C2_12	0.63	0.72	1594.086	22	-27.53	19	10	70.9	70.7
A2B3C2_13	0.6	0.69	1580.736	18	-23.75	18	11	70.7	69
A2B3C2_14	0.64	0.72	1587.935	20	-27.55	19	9	72	71.2
A2B3C2_15	0.6	0.69	1500.211	17	-22.5	20	11	72.8	67.4
A2B3C2_16	0.6	0.69	1494.192	19	-30.04	18	13	73.8	68.2
A2B3C2_17	0.62	0.71	1554.415	16	-23.03	19	10	72.3	71.6
A2B3C2_18	0.6	0.69	1354.87	22	-33.49	21	10	71.7	70.6
A2B3C2_19	0.61	0.7	1666.273	16	-24.46	18	11	71.4	70.6
A2B3C2_20	0.6	0.69	1525.943	17	-18.98	18	9	72.9	66.7
A2B3C2_21	0.64	0.72	1678.048	19	-24.81	17	11	71.7	71.7
A2B3C2_22	0.64	0.72	1394.159	16	-22.44	18	10	73.5	73.6
A2B3C2_23	0.62	0.71	1521.908	19	-27.53	20	12	71.6	70
A2B3C2_24	0.63	0.71	1777.792	17	-23.75	19	9	71.2	69.8
A2B3C2_25	0.6	0.69	1652.654	19	-29.98	19	11	71.8	69.5
A2B3C2_26	0.6	0.69	1596.079	18	-26.15	20	12	71.5	68.7
A2B3C2_27	0.6	0.69	1561.741	15	-20.11	17	13	70.5	66
A2B3C2_28	0.62	0.71	1391.571	21	-33.92	17	10	70.6	72.6
A2B3C2_29	0.6	0.69	1364.3	21	-31.15	20	10	72.3	70.1
A2B3C2_30	0.62	0.71	1646.259	20	-30.44	19	10	70.5	69.9
A2B3C2_31	0.6	0.7	1524.079	16	-25.45	18	12	73.1	70.4
A2B3C2_32	0.63	0.71	1725.014	16	-25.93	19	14	71.8	70
A2B3C2_33	0.6	0.69	1401.853	17	-25.45	17	8	72.7	70.1
A2B3C2_34	0.64	0.72	1423.633	15	-20.43	19	9	71.5	71.5
A2B3C2_35	0.64	0.72	1710.533	18	-23.87	21	11	71.9	71.6
A2B3C2_36	0.6	0.69	1393.989	20	-31.56	20	15	69.9	70
A2B3C2_37	0.6	0.69	1439.228	17	-26.01	20	11	72.4	67.9
A2B3C2_38	0.6	0.69	1757.173	17	-23.9	19	10	72.2	67.6
A2B3C2_39	0.62	0.7	1506.595	17	-25.02	19	11	74.7	71.3
A2B3C2_40	0.6	0.69	1612.364	19	-26.2	18	11	72.4	69

A2B3C2_41	0.59	0.69	1485.701	20	-33.98	18	10	72.7	68.4
A2B3C2_42	0.61	0.7	1463.443	20	-25.09	20	11	71.7	68.6
A2B3C2_43	0.63	0.72	1518.177	11	-14.99	19	11	72.1	72.2
A2B3C2_44	0.59	0.68	1507.327	13	-15.97	19	11	72.7	68.9
A2B3C2_45	0.6	0.69	1359.571	16	-22.96	17	12	72.4	69.5
A2B3C2_46	0.6	0.69	1438.948	20	-28.27	19	13	72.1	68.6
A2B3C2_47	0.6	0.69	12225.364	19	-26.62	18	11	72.9	69.6
A2B3C2_48	0.6	0.69	1473.016	15	-22.66	19	12	71.7	66.7
A2B3C2_49	0.62	0.71	1769.41	17	-24.97	18	12	71.7	69.9
A2B3C2_50	0.59	0.69	1360.698	18	-22.35	19	10	72.4	67.9
A2B3C2_51	0.63	0.72	1440.986	16	-25.01	18	10	71.2	72.1
A2B3C2_52	0.63	0.72	1444.82	20	-27.08	20	11	72.7	70.4
A2B3C2_53	0.64	0.72	1629.793	21	-25.89	19	15	72.6	70.6
A2B3C2_54	0.58	0.68	1534.85	19	-22.54	19	11	72.8	69
A2B3C2_55	0.59	0.68	1532.948	21	-26.93	20	10	72.9	70.6
A2B3C2_56	0.59	0.68	747.409	19	-28.36	19	10	72	68.5
A2B3C2_57	0.6	0.69	1297.992	21	-28.79	21	11	72.7	71.2
A2B3C2_58	0.6	0.7	1687.079	15	-21.82	20	10	72	70
A2B3C2_59	0.61	0.7	1592.092	22	-29.56	20	10	72.5	67.7
A2B3C2_60	0.59	0.69	1405.245	19	-29.09	19	10	71.7	69
A2B3C2_61	0.61	0.7	1382.779	22	-15.21	20	11	74.6	70.5
A2B3C2_62	0.6	0.7	1305.295	19	-25.06	16	14	72.9	73.1
A2B3C2_63	0.62	0.71	1519.127	16	-18.34	21	14	71.2	70.1
A2B3C2_64	0.63	0.72	1283.134	17	-27.98	21	11	72	70.6
A2B3C2_65	0.6	0.69	1645.243	16	-33.02	21	10	73.1	67.7
A2B3C2_66	0.6	0.69	1498.399	18	-19.85	19	11	72.9	71.4
A2B3C2_67	0.61	0.7	1529.829	13	-28.98	21	11	73.9	70
A2B3C2_68	0.6	0.69	1302.069	14	-15.42	20	10	72.5	69.8
A2B3C2_69	0.64	0.72	1727.768	20	-24.37	19	10	70.6	70.3
A2B3C2_70	0.61	0.7	1530.766	22	-21.34	18	8	74.3	67
A2B3C2_71	0.6	0.69	1749.75	12	-15.11	21	9	71.9	67.1
A2B3C2_72	0.6	0.69	1705.893	11	-27.9	21	11	71.3	73.1
A2B3C2_73	0.6	0.69	1647.667	14	-19	18	14	72.4	73.3
A2B3C2_74	0.63	0.72	1426.384	17	-26.53	20	10	71.2	70.9
A2B3C2_75	0.61	0.7	1348.308	19	-29.9	16	10	74.3	73.3
A2B3C2_76	0.59	0.69	1726.666	18	-15.92	17	11	73.4	68.7
A2B3C2_77	0.59	0.69	1708.957	12	-27.62	16	15	72.3	66.3
A2B3C2_78	0.61	0.7	1375.462	15	-24.45	21	8	72.3	70.2
A2B3C2_79	0.64	0.73	1545.285	18	-19.89	20	12	72.2	72.3
A2B3C2_80	0.59	0.69	1600.62	19	-20.3	19	8	73	71.6
A2B3C2_81	0.59	0.68	1305.781	13	-26.92	20	11	72	70.9
A2B3C2_82	0.62	0.7	1352.164	11	-30.97	20	11	72.5	66.4
A2B3C2_83	0.63	0.72	1612.757	16	-19.9	19	9	71.2	71.4
A2B3C2_84	0.6	0.7	1448.362	14	-32.95	18	15	74.6	67.8
A2B3C2_85	0.61	0.7	1372.005	20	-21.62	18	10	74.4	71.4
A2B3C2_86	0.64	0.72	1623.726	19	-22.46	20	10	72.6	71.4

A2B3C2_87	0.63	0.72	1607.436	16	-23.84	19	12	70.7	71.3
A2B3C2_88	0.59	0.68	1661.649	16	-33.22	20	9	73.9	66.4
A2B3C2_89	0.6	0.69	1690.222	21	-29.65	20	15	70.5	72.4
A2B3C2_90	0.62	0.7	1461.978	17	-25.29	18	13	72.6	73.5

Extended Data Table 3. AlphaFold3 predictions of A2B3C3 (ipTM > 0.66) and A2B3C2 (ipTM > 0.68) used for virtual complex model construction.

	BSA (Å²)	Hydrogen Bond Analysis		hydrophobic pairing	
		Effective Bonds	Average Energy	mimic with IL-6	mimic with sIL-6R
A2B3C3	1801.79	13	-12.19	18	23
A2B3C2	1759.434	13	-17.82	19	10
A3B2C2	1301.479	28	-40.46	-	41

Extended Data Table 4. Interface hotspot residues (Z-score > 2.00) identified by contact-based energetic scoring.

Model	Residue	Chain	Score	Z-score
top1	ARG6	IL-6	92.5	2.46
top1	TYR13	IL-6	91	2.4
top1	TRP169	sIL-6R	575.5	6.07
top1	ARG115	sIL-6R	251.5	2.32
top1	TYR50	mimic	118	3.47
top1	TYR25	mimic	82	2.04
top1	ARG217	mimic	223	1.99
top2	TYR13	IL-6	96	2.42
top2	ARG6	IL-6	93	2.31
top2	TRP154	sIL-6R	100	2.05
top2	ARG136	mimic	175.5	4.33
top2	TYR50	mimic	127	3.58
top2	TYR25	mimic	87.5	2.1
top3	ARG147	sIL-6R	305	3.79
top3	ARG115	sIL-6R	304	3.77
top3	ARG173	sIL-6R	212.5	2.34
top3	TRP154	sIL-6R	204.5	2.21
top3	TYR110	mimic	232.5	2.65
gp130	TYR31	IL-6	65.5	2.74
gp130	PHE169	sIL-6R	62	2.54
gp130	TYR168	sIL-6R	54	2.07
gp130	ASP262	mimic	76	2.99

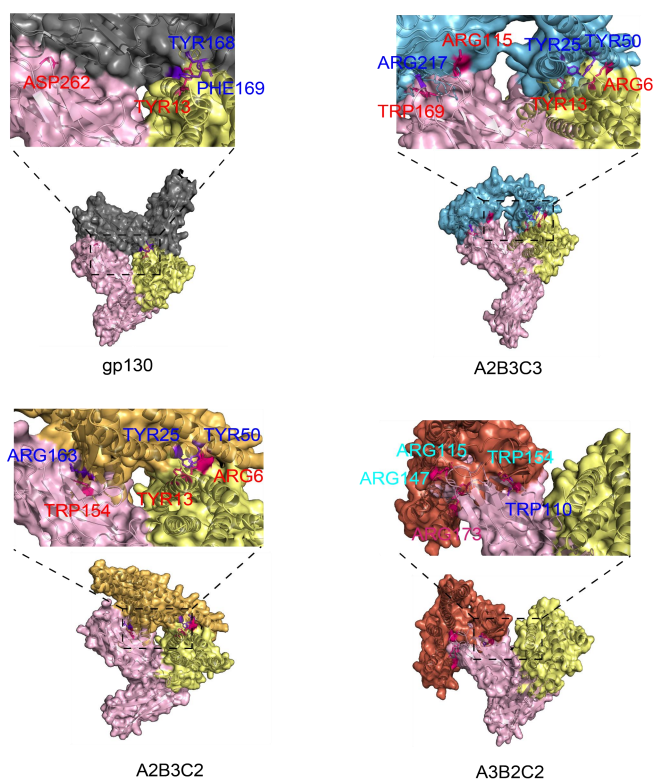


Figure 1. Structural models predicted by AlphaFold3 for the IL-6/sIL-6R α complex bound to gp130 or the indicated designed proteins.

6. AI Use Statement:

AI tools were used to assist with AlphaFold code generation and debugging. The team reviewed and tested the code. All results are true results. Computational prediction generated using AI, not experimentally validated by wet experiment. We employed a combined approach of computational design and experimental verification. Therefore, for results from AI-assisted prediction/simulation, we will continue to conduct wet experimental verification to confirm their validity.